

3.4 Phonological Scansion theory

3.4.1 Phonological Scansion Procedure

I have synthesised the phonological scansion procedure from the theory outlined by Hayes and Kaun (1996), based on the idea that only the ends of phonological units are significant for meter assignment, and that larger phonological units have greater weight in this assignment – Table 29 gives an analysis of one line (further details are given in the Appendices on p. 148).

Type	Analysis		
“Shall I compare thee to a summer’s day?” (Shakespeare, Sonnet 130.1)			
Stresses	Shall(m) I(n) com(w)pare(s) thee(n) to(m) a(w) sum(s)mer’s(w) day(s)?		
Phonological Analysis	<{[Shall I /compare\ thee]} {[to a /summer’s\] [day?]}>		
Key Stresses	<{[xx/ws\n]}{[xx/sw\]s]}>		
Phonological Scansion	10 syllables, 4 main stresses <u>Duple units:</u> Iambic (ws) = final IU (‘mer’s day’), 1PW2 (‘a sum’), 1PW2 (‘compare’) [total=5] Trochaic (nx) =1PP (‘thee to’) [total=2] <u>Triple units:</u> Amphibrachic (wsw) = 1PW2 (‘a summer’s’) [total=1] Dactylic (sww) =1PW4 (‘Shall I com’), 1PW4 (‘pare thee to’) [total=4]		
Analysis	The line matches 4/5 weighted units with a dactylic rhythm, but one unit is not matched (triple rhythms require 100% match): the line is not dactylic. The line is iambic with 5/7 weighted units matched (71%)		
Result	The analysis indicates that the line is iambic pentameter (mnlwslnmlwslws). The iambic rhythm is strong (71% consistent).		
Analysis Symbol	Scansion Symbol	Meaning	Scansion Weighting
/\	PW	Polysyllabic Word (incl. modal verb phrases)	1
[]	CP	Clitic Phrase	1
{ }	PP	Phonological Phrase	2
< >	IU	Intonational Unit	3

Table 29: Phonological Scansion of Shakespeare, Sonnet 130.1

3.4.2 Theory: Accuracy against Experts

(the results are given in 4.2.1)

In order to assess the effectiveness of the theory, I have collated a test corpus which includes the examples matching phonological phrases to meter given by Hayes and Kaun (1996) and Hayes (1984a), as well as examples of expert scansions of 17 individual verses and 32 poems – 372 verses in all. I can therefore determine whether the theory's output approximates expert scansion better than the scansions produced by the systems of Plamondon, Hartman and Fabb. The expert scansion used has been regularised by principles agreed in Baker (1996) to screen out idiosyncrasies. The sample size and the method are largely the same as those used by Hayes (2005) to test his textsetting algorithm allowing a comparison with his results.

Category	Description
Corpus	Poetry corpus of 372 verses, including 313 iambic poems (to assess Scandroid) and Shakespeare's Sonnet 130 (to assess AnalysePoems)
Test Subjects	1 test subject
Success Rate	Categories 1-6, 100%, categories 7-8 80%. Improves on competing theories' success rates or analyses, eg/ Hayes (71%), Fabb, Hartman, Plamondon, Groves
Tests Objectives	T1, T2, T3, T4
Method	The Phonological Scansion procedure is processed manually, by a human being. Results are compared to expert assessments and to competing procedures' results for verses in all eight categories of difficulty described above. The scansions will also be compared to non-expert scansions where these are available to determine if any approximate non-expert assumptions.
References	Hayes(1984), Hayes and Kaun (1996), Hayes (2005)
Assessment	The results are analysed for statistical confidence, and compared to each other. The iambic poems are used to assess Scandroid's performance, Sonnet 130 assesses AnalysePoems. Each assessment is presented in a separate results table, including the assessment of the entire corpus.

3.4.3 Phonological Scansion Application

The Phonological Scansion Theory is implemented in the Calliope application (the program flow is given in Figure 4).

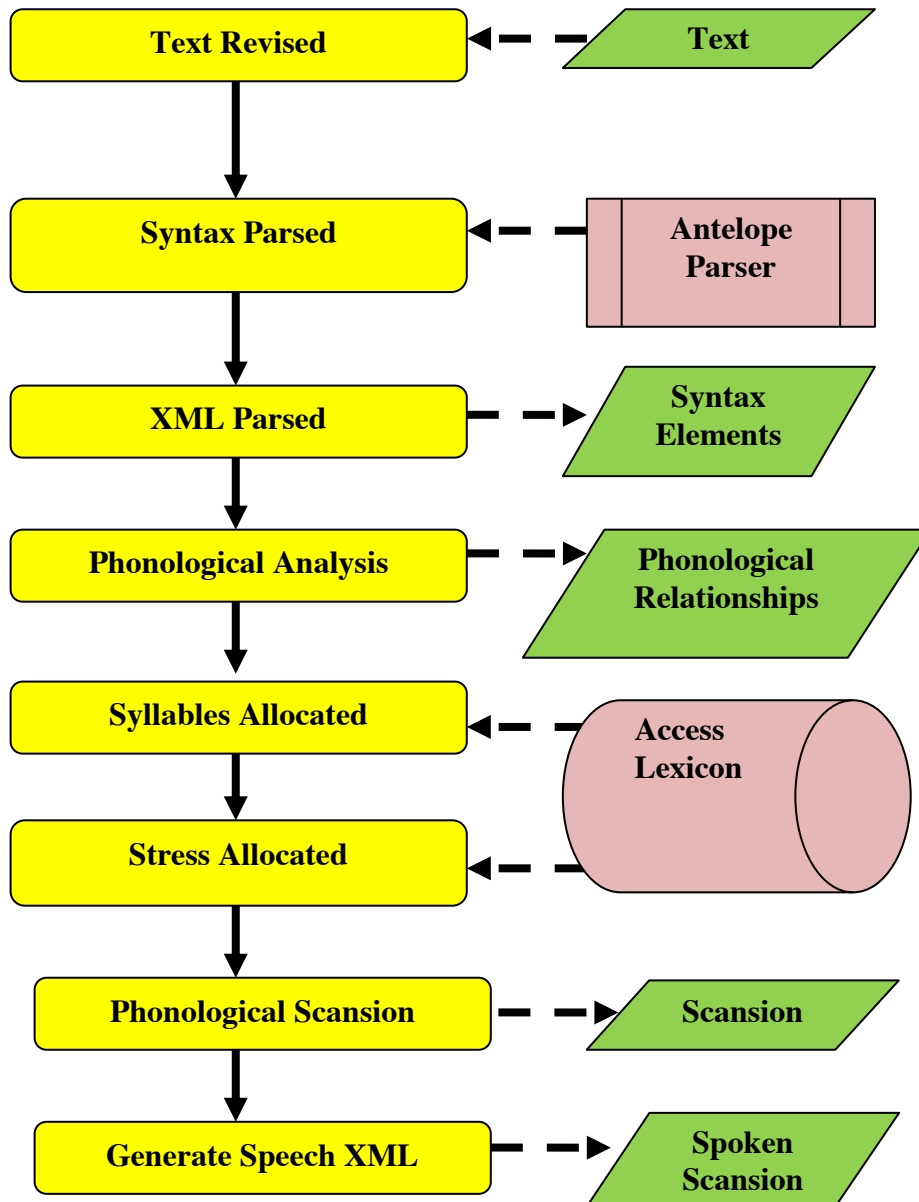


Figure 4: Calliope application program flow

An Antelope Natural Language Processor parses text into syntactic elements and relationships. Using this analysis, words are assigned stress from a lexicon, and are grouped into phonological units. The output is processed by a Phonological Scansion

algorithm, and the result is assessed for consistency: if the rhythm does not meet an arbitrary consistency, variant scansion of the line are assessed to determine the most consistent – see Table 30.

Type	Data
Original Text	Shall I compare thee to a summer's day?
Revised Text	Shall I compare you to a summer's day?
Syntactic analysis	Shall/VBD I/PRP compare/VB you/PRP to/TO a/DT summer/NN 's/POS day/NN ? (SQ Shall (NP I) (VP compare (NP you) (PP to (NP (NP a summer 's) day))) ?)
Phonological analysis	<{[Shall I compare thee]} {[to a summer's] [day]}>
Stress Patterns	shall(1) I(1) Com(0)pare(2) thee(1) To(1) A(1) Sum(2)mer(0)'s(-1) Day(2) nmwsnmnswm
All Stresses	<{[nm/ws\n]}mn/sw\m]>
KeyStresses	<{[xx/ws\n]}{[xx/sw\s]}>
Meter	iambic pentameter (from the end back)
Stresses	(10 overall, 3 strong)
Analysis	Iamb(Reverse):5, 1IU, 2PW Trochee(Reverse):2, 1PP Dactyl(Reverse):3, 1IU Anapaest(Reverse):2, 1PW
Scansion	xxlwslnx\lxlwm
Certainty	71% (5/7). The threshold is 55%, so no variants are tested.
Conclusion	The meter is iambic pentameter with a consistency of 71%

Table 30: Calliope analysis of Shakespeare, Sonnet 130.1

3.4.4 Application: Accuracy against Theory

(the results are given in 4.2.3)

A second experiment compared the application's phonological scansion with the theory's scansion for a corpus of test verses to determine how closely the application reproduces the theoretical results.

Category	Description
Corpus	Poetry corpus of 148 verses previously analysed by the Phonological Scansion theory, and representing the 8 categories of difficulty mentioned above
Test Subjects	Computer
Success Rate	Computer scansions match theoretical scansions 90% of time overall, and 100% of the time for categories 1 and 2.
Method	The computer will process the corpus using the application's Phonological Scansion, and compare to the Phonological Scansion's theoretical output.
References	Stauder (2000)
Assessment	The results are analysed for statistical confidence. The results are analysed within each difficulty category, and overall.

3.5 Phonological Scansion Application, Calliope

The main point of the research is to evaluate Phonological Scansion against Scandroid, using both Scandroid's native stress assignment system and the linguistic stress assignments.

The data for Scandroid's system is taken directly from the Scandroid program. In order to replicate the effects of scanning the revised stresses, I have reduced the four levels of stress to two levels depending on the relative weights of the stresses within a foot. I have then used words which Scandroid consistently scans as either strong ("test") or weak ("or") to reproduce the binary stress pattern, and had Scandroid scan these.

The reduction to binary stresses oversimplifies the procedure for Scandroid, but this is unavoidable – whilst ambiguous stresses could have been used for secondary and relatively weak stress (m and n), two consecutive ambiguous stresses in a foot (e.g. mn) might have been processed with incorrect relative stress whereas it is acceptable to most metrists to use relative stresses within feet.

3.5.1 Application: Speed against Scandroid

(the results are given in 4.3.2)

One criterion for success is efficiency, so the systems scanned the same verses on the same computer and the processing speed was recorded. The verses consisted of five iambic pentameter verses (since Scandroid copes best with iambic pentameter), of varying degrees of difficulty. Logs were used to indicate how long the program stayed in key functions. This quantitative data was assessed to determine whether the phonological scansion functions speeded up the scansion process or not. This test is similar to Zhang and Cercone (1999) who record the time taken for a CPU to process a stress assignment algorithm.

Speed tests were performed with Scandroid by processing large quantities of identical text, recording the time taken and calculating an average. Although it would have been preferable to use a computer timer procedure, C# versions of Scandroid proved unreliable, and the original Python program did not allow modifications. Moreover, since Scandroid is not consistent in the results it produces for a single text, an average at least calculates the most likely output.

Category	Description
Corpus	Poetry corpus of 5 iambic pentameter verses
Test Subjects	Computer
Success Rate	Calliope faster than Scandroid
Tests Objectives	A2
Method	the computer processes the corpus using the application's Phonological Scansion, and also using Scandroid with its own stress assignments. The speed of each of the processes is compared for the items in the corpus.
References	Zhang & Cercone (1999)
Assessment	The average speed over the 5 verses is compared to determine the faster application

3.5.2 Application: Accuracy against Scandroid

(the results are given in 4.3.1)

This test assessed how accurately the three systems replicate reliable expert scansion. The test was restricted by the features of the Scandroid program: the corpus was composed of representative verses of each common variant of iambic, anapaestic and trochaic verse, as well as some free verses – although the corpus was small, increasing it would not have extended the scope of the test though it might have increase the accuracy slightly. The three types of data which Scandroid displays are collected and compared against expert scansion. The output accurately replicated expert scansion if it matched the expert in each type of data.

Since the test assessed whether Calliope's stress assignments improved Scandroid's accuracy, this is a key test for evaluating the central proposition of the project – that stress assigned with reference to syntax improves scansion.

Category	Description
Corpus	Poetry corpus of 32 verses, each representing a common variant of anapaestic or iambic meters
Test Subjects	Computer
Success Rate	Calliope is more accurate than Revised Scandroid, if not Scandroid. Revised Scandroid is more accurate than Scandroid
Tests Objectives	A1, A3, B1, C2
Method	The computer processes a corpus of iambic pentameter and anapaestic meter using the application's Phonological Scansion, Scandroid and Revised Scandroid. The accuracy of each of the processes is compared for the items in the corpus against expert scansions for each meter. Data is collected for accuracy in identifying the correct number of feet, correct types of feet, and correct assignment of meter.
Assessment	The results are analysed for statistical confidence, and broken down by meter type.

3.5.3 Application: Accuracy against Non-Experts

(the test document is in the Appendices on pp. 200-203, and results are given in 4.4.3)

The output from the three systems was also compared to scansion produced by human beings to evaluate which is most effective. Although it was primarily compared to scansion produced by experts (who are more likely to identify the intended rhythm), since scansion is to some degree subjective, I also had non-experts scan verses to evaluate scientifically whether the systems are adopting non-expert assumptions in the scansion procedure.

For simplicity and consistency, these three tests were run on the same data which consisted of poems for which I have found expert scansions and which could be interpreted as iambic pentameter (that is, in meters with nine to twelve syllable patterns). This restriction tested Scandroid fairly since it only scans for ten syllable verse and is biased towards iambics.

There were at least 10 subjects for the non-expert test which was run on a computer (where subject chose the appropriate answer from drop-down lists or option buttons) or in hard-copy (where subjects marked the appropriate option from a list of alternatives) – this allowed subjects to use the format they were most comfortable with, so that they could focus on the test. Subjects were chosen with a variety of experience in poetry analysis to control against inexperienced subjects being led by external factors (such as question implication). In order to determine their relative experience, they were asked to assess their own experience and give details of their education background and exposure to poetry.

The subjects were given written verses with scansion produced by Scandroid with and without revised stress assignment, the phonological scansion procedure and other